

Clinical Trial Data Cleaning and Quality Assessment of Multisite Clinical Study Data

Project Summary

This project involved cleaning and quality assessment of multisite clinical trial datasets collected across five study sites. The datasets contained inconsistent categorical entries, mixed date formats, non-standard measurement units, typographical errors, ambiguous dates, and identifier discrepancies.

Using R, a reproducible data cleaning workflow was developed to standardize variables, validate identifiers, harmonize units, identify data quality issues, and generate analysis-ready datasets. The project produced cleaned datasets for patients, adverse events, laboratory results, and dosing records while documenting records requiring further review.

1. Introduction

Clinical trial datasets are often collected from multiple sites and may contain inconsistencies, missing values, duplicate formats, invalid identifiers, ambiguous dates, and non-standard variable entries. Poor data quality can negatively affect statistical analysis, regulatory reporting, and study conclusions.

This project focused on cleaning and standardizing multisite clinical trial datasets using R. The project involved patient records, adverse events, laboratory results, dosing records, and site-level data. The goal was to improve data consistency, identify data quality issues, and produce analysis-ready datasets.

2. Objectives

The objectives of this project were to:

- i. Import and integrate multisite clinical datasets into R.
- ii. Standardize categorical variables across datasets.
- iii. Clean and harmonize dates, units, and identifiers.
- iv. Detect inconsistencies and potential data entry errors.
- v. Generate cleaned datasets for downstream analysis.
- vi. Document quality issues identified during the cleaning process.

3. Datasets

Five datasets were used in this project.

Dataset	Description	Records	Variables
Patients	Demographic and enrollment information	141	12
Adverse Events	Safety and adverse reaction records	215	10

Dosing	Treatment administration records	307	10
Laboratory Results	Clinical laboratory measurements	410	10
Sites	Study site information	5	6

4. Packages Used

The entire workflow was implemented in R.

Package	Purpose
tidyverse	Data manipulation and transformation
lubridate	Date parsing and standardization
janitor	Data cleaning utilities
stringi	Text normalization
rio	Data import and export
skimr	Data quality assessment

5. Data Cleaning Workflow

5.1 Patient Data Cleaning

The patient dataset contained demographic and enrollment information. Several variables required standardization and validation.

Demographic Variable Standardization

The following variables were standardized:

- Sex
- Ethnicity
- State of Origin
- Treatment Arm
- Study Status

Examples of corrections include:

Original Entry	Standardized Value
F	Female
Woman	Female

M	Male
Haoussa	Hausa
Ibo	Igbo
Control	Placebo

This process ensured consistent coding across all patient records.

Weight Standardization

Patient weight was recorded using different units and formats. All weights were converted to kilograms.

Examples:

Original Value	Standardized Value
150 lb	68.0 kg
70 kg	70.0 kg

Height Standardization

Height values were recorded using centimeters, inches, and feet/inches combinations. All measurements were converted to centimeters.

Examples:

Original Value	Standardized Value
175 cm	175 cm
68 in	172.7 cm
5'8	173 cm

BMI Calculation

Body Mass Index (BMI) was calculated using standardized weight and height measurements according to:

$$BMI = \frac{Weight (kg)}{Height (m)^2}$$

Date Standardization

Date of birth and consent dates were recorded in multiple formats. These dates were standardized into a consistent date format.

Examples of formats handled include:

- YYYY-MM-DD
- DD/MM/YYYY

- Month DD, YYYY
- Excel serial dates

Future birth dates resulting from century-entry errors were adjusted accordingly.

Site Identifier Standardization

Site identifiers were harmonized to a single naming convention.

Examples:

Original	Standardized
ST01	SITE01
S01	SITE01
SITE01	SITE01

5.2 Adverse Events Cleaning

The adverse events dataset contained information related to treatment safety and adverse reactions.

Relationship Standardization

Different representations of treatment relationship were harmonized.

Original	Standardized
Y	Yes
Related	Yes
Possibly Related	Yes
N	No
Unrelated	No

Seriousness Standardization

Values representing seriousness were standardized to either:

- Yes
- No

Outcome Standardization

Outcome categories were standardized into:

- Resolved
- Not Resolved
- Ongoing

- Unknown
- Death

Severity Standardization

Severity grades were harmonized into:

- Grade 1
- Grade 2
- Grade 3
- Grade 4

Date Validation

Onset and resolution dates were standardized and checked for ambiguity.

Dates such as:

01/02/2025

were flagged because they could represent:

- 1 February 2025
- 2 January 2025

Such records were exported for manual review.

Identifier Validation

Subject IDs and adverse event IDs were standardized and checked against expected ranges.

Records containing IDs beyond expected limits were flagged as potential invalid identifiers.

5.3 Laboratory Result Cleaning

The laboratory dataset contained repeated measurements collected during follow-up visits.

Visit Standardization

Visit names were harmonized into a consistent study schedule.

Examples:

Original	Standardized
BL	Baseline
V0	Baseline

Week4	Visit 1
EOS	Final Visit

Sample Date Standardization

Sample collection dates were standardized into a consistent format.

Ambiguous dates were identified and flagged for manual review.

Laboratory Result Extraction

Some result fields contained text combined with numeric values.

Examples:

Original	Standardized
13.5 g/dL	13.5
145 mg/dL	145

Numeric values were extracted and converted into analysis-ready form.

Unit Harmonization

Laboratory units were standardized.

Examples:

Original	Standardized
IU/L	U/L
Units/L	U/L
G/L	g/dL

Where necessary, results were converted to equivalent units to ensure consistency.

Identifier Validation

Laboratory identifiers were checked for potential invalid values and exported for review when necessary.

5.4 Dosing Dataset Cleaning

The dosing dataset contained treatment administration information.

Route Standardization

Administration routes were harmonized.

Original	Standardized
IV	Intravenous
Infusion	Intravenous
PO	Oral
SC	Subcutaneous

Frequency Standardization

Dosing frequencies were standardized.

Original	Standardized
BID	Twice/day
QD	Once/day
QW	Once/week

Dose Modification Standardization

Dose modification indicators were converted into:

- Yes
- No

Date Standardization

Treatment dates, planned start dates, and planned end dates were converted into a common date format.

Logical Consistency Checks

Records where the planned end date occurred before the planned start date were identified and exported for review.

6. Data Quality Issues

The following data quality issues were observed across the datasets.

Issue Category	Description
Typographical Errors	Misspelled categories and inconsistent spellings
Mixed Date Formats	Multiple date representations
Ambiguous Dates	Dates with unclear day/month ordering
Mixed Measurement Units	Pounds, kilograms, inches, and centimeters
Invalid Identifiers	IDs outside expected ranges

Logical Inconsistencies	End dates before start dates
Non-standard Categories	Multiple representations of the same category

Result Summary

The data cleaning process successfully standardized demographic variables, treatment information, laboratory measurements, adverse event records, and dosing information across all study datasets.

Several data quality issues were identified during the review process, including typographical errors, mixed measurement units, inconsistent categorical coding, ambiguous dates, invalid identifiers, and logical inconsistencies in dosing schedules. Records requiring manual review were isolated and exported separately to facilitate further quality assurance activities.

The final outputs consisted of cleaned datasets suitable for downstream statistical analysis and reporting, together with supplementary files documenting unresolved data quality concerns.

7. Outputs Generated

The following outputs were produced during the project.

Output File	Description
cleaned_patients_data.xlsx	Cleaned patient dataset
cleaned_adverse_event_data.xlsx	Cleaned adverse event dataset
ghost_ids_adverse_data.xlsx	Potential invalid identifiers
ambiguous_dates_in_adverse_data.xlsx	Ambiguous dates requiring review
cleaned_lab_result_data.xlsx	Cleaned laboratory dataset
ghost_lab_id.xlsx	Potential invalid laboratory IDs
cleaned_dosing_data.xlsx	Cleaned dosing dataset
inco_start_end_trt_date.xlsx	Dosing date inconsistencies

8. Key Skills

This project demonstrates practical skills in:

- Clinical Data Cleaning
- Clinical Data Management
- Data Validation

- Data Query Identification
- Data Quality Assessment
- Data Quality Control (QC)
- Date Standardization
- Unit Harmonization
- Identifier Verification
- Data Transformation Using R
- Reproducible Workflows
- Automated Data Export
- Clinical Research Data Handling

9. Challenges Encountered

Several challenges were encountered during the cleaning process.

- Multiple dates formats across datasets.
- Ambiguous date entries.
- Inconsistent spellings and category labels.
- Mixed measurement units.
- Invalid identifiers.
- Logical inconsistencies in dosing records.
- Missing and incomplete records.

These issues required the implementation of validation checks and standardization rules to ensure data quality.

10. Conclusion

This project demonstrated the end-to-end cleaning and quality assessment of multisite clinical trial data using R. Through systematic standardization of categorical variables, harmonization of units, date validation, identifier verification, and logical consistency checks, the datasets were transformed into analysis-ready formats.

The workflow also identified several important data quality issues requiring manual review, including ambiguous dates, potential invalid identifiers, and inconsistent data entry practices. The project highlights the importance of reproducible data cleaning pipelines in clinical research and demonstrates practical clinical data management skills applicable to real-world studies.